

BUSINESS REQUIREMENT DOCUMENT (BRD)

AI Pulse 2030

Global AI Ecosystem Analytics Platform

End-to-End Data Analytics Portfolio Project

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Prepared For	FutureAI Analytics Pvt. Ltd.(Functional/Imaginary)
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1. Executive Summary

Artificial Intelligence (AI) is transforming education, technology, and the global workforce by enabling automation, improving productivity, and supporting data-driven decision-making. As AI adoption continues to grow, organizations require centralized analytics platforms capable of converting large volumes of data into meaningful business insights.

AI Pulse 2030 – Global AI Ecosystem Analytics Platform is an end-to-end Data Analytics portfolio project that integrates multiple AI-related datasets to analyze student AI adoption, AI tools, GitHub repository statistics, and workforce trends. The project follows a structured analytics workflow using Microsoft Excel for data cleaning, PostgreSQL and SQL for database management and business analysis, and Microsoft Power BI for interactive dashboard development.

The solution provides decision-makers with a centralized platform to monitor AI adoption, evaluate workforce trends, analyze AI technologies, and support strategic planning through interactive visualizations and key performance indicators. This Business Requirement Document defines the business requirements, project scope, implementation approach, and expected deliverables while demonstrating professional business analysis practices.

2. Business Problem

Organizations often rely on multiple disconnected datasets to analyze Artificial Intelligence trends, resulting in fragmented reporting and inefficient decision-making. Educational institutions require insights into student AI adoption, technology companies need visibility into AI tool growth, and employers seek workforce intelligence regarding future demand, automation risks, and emerging AI skills.

The lack of a unified analytics platform makes it difficult to compare trends, identify opportunities, and generate actionable business insights. Manual data preparation also increases reporting time and reduces analytical efficiency.

The **AI Pulse 2030** platform addresses these challenges by integrating multiple datasets into a centralized Business Intelligence solution. Through data cleaning, SQL analysis, and interactive Power BI dashboards, the platform enables stakeholders to monitor AI adoption, evaluate workforce transformation, and explore the global AI ecosystem using reliable, data-driven insights.

3. Business Objectives

The AI Pulse 2030 project aims to develop an end-to-end Business Intelligence solution that transforms AI-related datasets into meaningful business insights.

Objectives:

- Build an analytics solution using Excel, PostgreSQL, SQL, and Power BI.

- Analyze student AI adoption and academic trends.
- Evaluate AI tools, GitHub activity, and technology adoption.
- Assess workforce demand, salaries, and AI replacement risk.
- Design interactive dashboards with business KPIs.
- Support data-driven decision-making through visualization and reporting.

4. Stakeholders

Stakeholder	Role
Executive Management	Strategic decision-making
Business & Data Analysts	Data analysis and reporting
HR Teams	Workforce planning
Educational Institutions	Student AI analysis
Technology Companies	AI market insights
Students & Researchers	Learning and research

5. Project Scope

In Scope

- Data cleaning in Microsoft Excel
- Database design and SQL analysis
- Interactive Power BI dashboards
- KPI and DAX development
- Business documentation

Out of Scope

- Real-time APIs
- Machine Learning
- Cloud deployment
- User authentication
- Mobile application
- Power BI Service

6. Data Sources

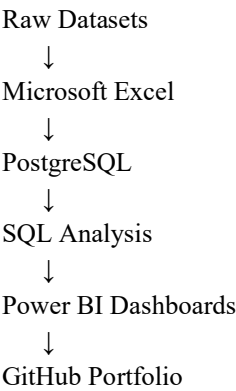
The project integrates three datasets to provide a comprehensive view of the AI ecosystem.

Dataset	Purpose
Student AI Adoption	Analyze AI usage, learning behavior, and academic trends.
AI Tools Dataset	Evaluate AI tools, companies, GitHub activity, and categories.
AI Impact on Jobs 2030	Analyze salaries, AI replacement risk, and future workforce demand.

7. Technology Stack

Technology	Purpose
Microsoft Excel	Data Cleaning
PostgreSQL	Database Management
SQL	Business Analysis
Power BI	Dashboard Development
GitHub	Version Control & Documentation

8. Project Workflow



9. Business Requirements

Microsoft Excel

- Import and clean raw datasets.
- Remove duplicates and missing values.
- Standardize data for analysis.

PostgreSQL & SQL

- Store cleaned datasets.
- Perform data modeling and SQL analysis.
- Generate business insights using queries.

Microsoft Power BI

- Create relationships and DAX measures.
- Build interactive dashboards and KPIs.
- Enable filtering and drill-down analysis.

10. Dashboard Overview

Dashboard	Purpose
Executive Dashboard	Overall project KPIs and insights
Student AI Dashboard	Student AI adoption analysis
AI Tools Dashboard	AI tools and GitHub analysis
Workforce Dashboard	AI impact on jobs and salaries
Executive Insights	Key findings and recommendations

11. Key Performance Indicators (KPIs)

Student Analytics

- Total Students
- Average GPA
- Average AI Usage
- Average Study Hours

AI Tools Analytics

- Total AI Tools
- Total GitHub Stars
- Total Forks
- Open Source %

Workforce Analytics

- Total Job Roles
- Average Salary
- Future Demand Score
- AI Replacement Risk

12. Expected Deliverables

The project will deliver the following outputs:

- Business Requirement Document (BRD)
- Cleaned Excel Datasets
- PostgreSQL Database
- SQL Scripts
- Power BI Dashboard (.pbix)
- Dashboard Screenshots
- GitHub Repository
- README Documentation
- Project Assets & Supporting Files

13. Assumptions

- Public datasets are accurate and suitable for analysis.
- Data is cleaned before loading into the database.
- Dashboard insights are based on sample datasets.
- Users have basic knowledge of Power BI reports.

14. Risks

- Incomplete or inconsistent source data.
- Static datasets may not reflect current trends.

15. Future Enhancements

- Live API integration
- Automated data refresh
- Cloud deployment
- Power BI Service publishing
- Predictive analytics
- AI-powered insights
- Mobile dashboard optimization

16. Success Criteria

The project will be considered successful if:

- Data is cleaned and validated successfully.
- SQL analysis generates accurate insights.
- Power BI dashboards are interactive and user-friendly.
- All business requirements are implemented.
- Documentation is complete and organized.
- GitHub repository is portfolio-ready.

17. References

- Kaggle Public Datasets
- Microsoft Excel Documentation
- PostgreSQL Documentation
- Microsoft Power BI Documentation
- GitHub Documentation

18. Author Information

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15. Disclaimer & Copyright

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